# 📗 Answer Key — XII Informatics Practices (065)

Unit Test – 2 (UT-2)

## Section A 1 mark each

| Q.No | Answer |
|---|---|
| 1 | b) INSTR() |
| 2 | c) Mesh |
| 3 | c) DAYNAME() |
| 4 | c) Router |
| 5 | c) TRIM() |
| 6 | b) 16 |
| 7 | b) COUNT(column_name) |
| 8 | d) WAN |
| 9 | b) ORDER BY |
| 10 | a) MOD() |
| 11 | c) UCASE('hello') |
| 12 | b) 2 |
| 13 | c) HAVING |
| 14 | b) To convert digital signals to analog signals and vice versa |

## Section B 2 marks each

### 15. A. ORDER BY vs GROUP BY

| ORDER BY | GROUP BY |
|---|---|
| Sorts rows in ascending/descending order. | Groups rows having same values in a column. |
| Used with SELECT to sort final result. | Used with aggregate functions (SUM, AVG, COUNT). |
| `SELECT * FROM EMP ORDER BY Salary DESC;` | `SELECT Dept, AVG(Salary) FROM EMP GROUP BY Dept;` |

### 15. B. PAN and LAN

- **PAN:** Network within a very small area (around a person), up to ~10 metres. *Example:* Mobile connected to laptop via Bluetooth.

- **LAN:** Network within a small area like office, school, or building. *Example:* Computers connected in a school computer lab.

**16. Output of SQL queries on PRODUCT**

*I.* `SELECT MAX(Price), MIN(Price) FROM PRODUCT;`

| MAX(Price) | MIN(Price) |
|---|---|
| 60000 | 500 |

*II.* `SELECT COUNT(*) FROM PRODUCT WHERE Qty > 20;`

```
2
```

**17. A. SQL query for current date and month name**

```
SELECT CURDATE(), MONTHNAME(CURDATE());
```

**17. B. Difference between Hub and Switch**

| Hub | Switch |
|---|---|
| Broadcasts data to all connected devices. | Sends data only to the intended destination device. |
| Works at physical layer. | Works at data link layer. |
| Half-duplex; causes collisions. | Full-duplex; reduces collisions. |
| Slower and less secure. | Faster and more secure. |

**18. SQL query on EMPLOYEE table**

*Query:* Display department-wise average salary, only for departments where average salary is greater than 60000.

```
SELECT Dept, AVG(Salary) AS Avg_Salary
FROM EMPLOYEE
GROUP BY Dept
HAVING AVG(Salary) > 60000;
```

## Section C

**19. Star Topology vs Bus Topology**

| Star Topology | Bus Topology |
|---|---|
| All nodes connected to a central hub/switch. | All nodes connected to a single backbone cable. |
| Failure of one node does not affect others. | Failure of backbone cable brings down entire network. |
| Easy to add/remove nodes. | Difficult to add/remove nodes. |
| Requires more cable. | Requires less cable. |
| Central device failure affects entire network. | No central device. |

*Star Diagram:*

```
      Node1
        |
Node2— HUB —Node3
        |
      Node4
```

*Bus Diagram:*

```
Node1   Node2   Node3   Node4
  |       |       |       |
==+=======+=======+=======+==   (Backbone cable)
```

### 20. SQL queries for string functions

```
-- I. Position of 'Science' in 'Computer Science'
SELECT INSTR('Computer Science', 'Science');

-- II. Length of 'Informatics Practices'
SELECT LENGTH('Informatics Practices');

-- III. First 5 characters from 'Database Management'
SELECT SUBSTRING('Database Management', 1, 5);
```

## Section D 4 marks each

### 21. SQL queries on EMPLOYEE and DEPARTMENT

*I. Employee name, department name, and salary using equi-join*

```
SELECT E.Emp_Name, D.Dept_Name, E.Salary
FROM EMPLOYEE E, DEPARTMENT D
WHERE E.Dept_ID = D.Dept_ID;
```

*II. Department name and total salary using natural join*

```
SELECT D.Dept_Name, SUM(E.Salary) AS Total_Salary
FROM EMPLOYEE E NATURAL JOIN DEPARTMENT D
GROUP BY D.Dept_Name;
```

*III. Employee name, department name, and joining month for IT department*

```
SELECT E.Emp_Name, D.Dept_Name, MONTHNAME(E.Join_Date)
FROM EMPLOYEE E, DEPARTMENT D
WHERE E.Dept_ID = D.Dept_ID AND D.Dept_Name = 'IT';
```

*IV. Department name, location, and number of employees with more than 2 employees*

```
SELECT D.Dept_Name, D.Location, COUNT(*) AS Emp_Count
FROM EMPLOYEE E, DEPARTMENT D
WHERE E.Dept_ID = D.Dept_ID
```

```
GROUP BY D.Dept_Name, D.Location
HAVING COUNT(*) > 2;
```

### 22. XYZ Tech Solutions – Network Design

*I. Most suitable department to install the server:*

```
IT Department (highest number of computers: 90)
```

*II. Cable layout between departments:*

```
HR ── Finance ── IT ── Logistics

HR–Finance: 50 m
Finance–IT: 60 m
IT–Logistics: 60 m
Total = 170 m
```

*III. Hardware device to connect computers within each department:*

```
Switch
```

*IV. Most suitable topology:*

```
Star Topology
```

### 23. Network devices and their functions

*I. Repeater:*

```
[Device A] ── [Repeater] ── [Device B]
```

*II. Router:*

```
[Network 1] ── [Router] ── [Network 2]
```

*III. Gateway:*

```
[Network A (Protocol X)] ── [Gateway] ── [Network B (Protocol Y)]
```

*IV. Switch:*

```
    [PC1]
      |
[PC2]—[Switch]—[PC3]
      |
    [PC4]
```

## Section E 5 marks each

### 24. SQL query outputs on STUDENT and MARKS

*I.* `SELECT Name FROM STUDENT WHERE Class = '12' ORDER BY Name ASC;`

| Name |
|---|
| Ankit Sharma |
| Rehan Khan |
| Rohan Verma |

*II.* `SELECT DISTINCT UCASE(Subject) FROM MARKS WHERE Score > 80;`

| UCASE(Subject) |
|---|
| MATHEMATICS |
| CHEMISTRY |
| BIOLOGY |
| COMPUTER SCIENCE |

*III.* `SELECT STUDENT.Name, MARKS.Subject, MARKS.Score FROM STUDENT, MARKS WHERE STUDENT.StudentID = MARKS.StudentID;`

| Name | Subject | Score |
|---|---|---|
| Ankit Sharma | Mathematics | 85 |
| Ankit Sharma | Physics | 78 |
| Priya Singh | Chemistry | 88 |
| Rohan Verma | Mathematics | 92 |
| Shreya Patel | Biology | 81 |
| Rehan Khan | Computer Science | 93 |
| Priya Singh | Mathematics | 76 |

*IV.* `SELECT STUDENT.Name, SUM(MARKS.Score) AS Total_Marks FROM STUDENT, MARKS WHERE STUDENT.StudentID = MARKS.StudentID GROUP BY STUDENT.Name HAVING COUNT(MARKS.Subject) > 1;`

| Name | Total_Marks |
|---|---|
| Ankit Sharma | 163 |
| Priya Singh | 164 |

*V.* `SELECT Subject, AVG(Score) AS Average_Score FROM MARKS GROUP BY Subject;`

| Subject | Average_Score |
|---|---|
| Mathematics | 84.33 |
| Physics | 78.00 |

| Subject | Average_Score |
|---|---|
| Chemistry | 88.00 |
| Biology | 81.00 |
| Computer Science | 93.00 |

**25. Shiva Corporation – Network Design (Pune Campus)**

*I. Most suitable building to install the server:*

```
Administration (highest number of computers: 120)
```

*II. Cable layout between buildings:*

```
Administration ── Sales (60 m)
Sales ── Development (50 m)
Development ── Support (45 m)

Total cable = 155 m
```

*III. Hardware device to connect computers within each building:*

```
Switch
```

*IV. High-speed wired link with Mumbai HQ:*

```
Optical Fibre Cable (OFC)
```

*V. Type of network connecting Pune campus to Mumbai HQ:*

```
WAN (Wide Area Network)
```